Writing Equations from Word Situations

Turning a situation into an equation: adding and removing, sharing and scaling, and two-step charges

Directions:

- For every problem: name the unknown with the letter given, write an equation that says what the situation says, then solve it. Show the equation in your work space — it is half the answer.
- Units are stated in each problem. Give the answer as a plain number in those units.
- A useful test: substitute your answer back into your equation and check that both sides come out equal.

1. Maya had s stickers. She was given 14 more, and then she had 32 stickers. Write an equation for s and solve it.

Answer: ________________

2. A bag holds m marbles. Six friends share the bag equally and each one receives 15 marbles. Write an equation for m and solve it.

Answer: ________________

3. A tank held w litres of water. After 26 litres drained out, 48 litres were left. Write an equation for w and solve it.

Answer: ________________

4. A taxi charges 3 dollars to start the trip plus 2 dollars for each kilometre travelled. One trip cost 19 dollars. Let k be the number of kilometres travelled.

(a) Write the equation for k in the work space, and copy it on this line: ________

(b) the number of kilometres travelled: ________

5. A rope r centimetres long is cut into 8 equal pieces, and each piece is 2.5 centimetres long. Write an equation for r and solve it.

Answer: ________________

6. A swim club charges 12 dollars to join plus 5 dollars each month. A member has paid 57 dollars in total. Write an equation for the number n of months and solve it.

Answer: ________________

7. The temperature was t degrees Celsius at noon. It fell 17 degrees during the afternoon and reached -4 degrees Celsius. Write an equation for t and solve it.

Answer: ________________

8. A gardener plants 6 rows with the same number of plants in each row, and then puts 4 extra plants in a pot. Altogether there are 58 plants. Let p be the number of plants in one row.

(a) Write the equation for p in the work space, and copy it on this line: ________

(b) the number of plants in one row: ________

9. A recipe uses 3 times as many grams of flour as grams of sugar, and the flour and the sugar together weigh 640 grams. Write an equation for the mass g of sugar in grams, and solve it.

Answer: __________________

10. A school play sold adult tickets at 8 dollars each and sold 60 child tickets at 5 dollars each. The ticket money came to 940 dollars altogether. Let a be the number of adult tickets sold.

(a) Write the equation for a in the work space, and copy it on this line: _________

(b) the number of adult tickets sold: _________

(c) Explain what each part of your equation stands for in the situation. Write this in the work space.